

Trabalho Mininet

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CRIANDO TOPOLOGIA DE 6 SWITCHES:

`sudo mn -c`

`sudo mn --topo=linear,6 --mac --link=tc,bw=25` # BW = 25 Mbps, endereços MAC padronizados, controlador padrão do Mininet

```
mininet@mininet-vm: ~  
mininet@mininet-vm:~$ sudo mn -c  
*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes  
killall controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflowd  
ovs-controller udpbwtest mnexec ivs 2> /dev/null  
killall -9 controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openfl  
owd ovs-controller udpbwtest mnexec ivs 2> /dev/null  
pkill -9 -f "sudo mnexec"  
*** Removing junk from /tmp  
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log  
*** Removing old Xl1 tunnels  
*** Removing excess kernel datapaths  
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'  
*** Removing OVS datapaths  
ovs-vsctl --timeout=1 list-br  
ovs-vsctl --if-exists del-br s1 -- --if-exists del-br s2 -- --if-exists del-br s  
3 -- --if-exists del-br s4 -- --if-exists del-br s5 -- --if-exists del-br s6  
ovs-vsctl --timeout=1 list-br  
*** Removing all links of the pattern foo-ethX  
ip link show | egrep -o '([_[:alnum:]]+-eth[[:digit:]]+)'  
( ip link del s1-eth2;ip link del s2-eth2;ip link del s2-eth2;ip link del s1-eth  
2;ip link del s2-eth3;ip link del s3-eth2;ip link del s3-eth2;ip link del s2-eth  
3;ip link del s3-eth3;ip link del s4-eth2;ip link del s4-eth2;ip link del s3-eth  
3;ip link del s4-eth3;ip link del s5-eth2;ip link del s5-eth2;ip link del s4-eth  
3;ip link del s5-eth3;ip link del s6-eth2;ip link del s6-eth2;ip link del s5-eth  
3 ) 2> /dev/null  
ip link show  
*** Killing stale mininet node processes  
pkill -9 -f mininet:  
*** Shutting down stale tunnels  
pkill -9 -f Tunnel=Ethernet  
pkill -9 -f .ssh/mn  
rm -f ~/.ssh/mn/*  
*** Cleanup complete.
```

```
mininet@mininet-vm: ~  
mininet@mininet-vm:~$ sudo mn --topo=linear,6 --mac --link=tc,bw=25  
*** Creating network  
*** Adding controller  
*** Adding hosts:  
h1 h2 h3 h4 h5 h6  
*** Adding switches:  
s1 s2 s3 s4 s5 s6  
*** Adding links:  
(25.00Mbit) (25.00Mbit) (h1, s1) (25.00Mbit) (25.00Mbit) (h2, s2) (25.00Mbit) (2  
5.00Mbit) (h3, s3) (25.00Mbit) (25.00Mbit) (h4, s4) (25.00Mbit) (25.00Mbit) (h5,  
s5) (25.00Mbit) (25.00Mbit) (h6, s6) (25.00Mbit) (25.00Mbit) (s2, s1) (25.00Mbi  
t) (25.00Mbit) (s3, s2) (25.00Mbit) (25.00Mbit) (s4, s3) (25.00Mbit) (25.00Mbit)  
(s5, s4) (25.00Mbit) (25.00Mbit) (s6, s5)  
*** Configuring hosts  
h1 h2 h3 h4 h5 h6  
*** Starting controller  
c0  
*** Starting 6 switches  
s1 s2 s3 s4 s5 s6 ... (25.00Mbit) (25.00Mbit) (25.00Mbit) (25.00Mbit) (25.00Mbit)  
(25.00Mbit) (25.00Mbit) (25.00Mbit) (25.00Mbit) (25.00Mbit) (25.00Mbit) (25.00M  
bit) (25.00Mbit) (25.00Mbit) (25.00Mbit) (25.00Mbit)  
*** Starting CLI:
```

INSPECIONAR INFORMAÇÕES DE REDE:

Dentro do prompt do Mininet:

Listar nós (hosts e switches): **nodes**

Ver conexões entre nós: **net**

Ver detalhes de endereços e interfaces: **dump**

Inspecionar informações específicas de um nó: **h1 ifconfig -a**

Para o switch 1 (s1): **s1 ifconfig -a**

```
mininet@mininet-vm: ~  
*** Starting CLI:  
mininet> nodes  
available nodes are:  
c0 h1 h2 h3 h4 h5 h6 s1 s2 s3 s4 s5 s6  
mininet> net  
h1 h1-eth0:s1-eth1  
h2 h2-eth0:s2-eth1  
h3 h3-eth0:s3-eth1  
h4 h4-eth0:s4-eth1  
h5 h5-eth0:s5-eth1  
h6 h6-eth0:s6-eth1  
s1 lo: s1-eth1:h1-eth0 s1-eth2:s2-eth2  
s2 lo: s2-eth1:h2-eth0 s2-eth2:s1-eth2 s2-eth3:s3-eth2  
s3 lo: s3-eth1:h3-eth0 s3-eth2:s2-eth3 s3-eth3:s4-eth2  
s4 lo: s4-eth1:h4-eth0 s4-eth2:s3-eth3 s4-eth3:s5-eth2  
s5 lo: s5-eth1:h5-eth0 s5-eth2:s4-eth3 s5-eth3:s6-eth2  
s6 lo: s6-eth1:h6-eth0 s6-eth2:s5-eth3  
c0  
mininet> dump  
<Host h1: h1-eth0:10.0.0.1 pid=4528>  
<Host h2: h2-eth0:10.0.0.2 pid=4530>  
<Host h3: h3-eth0:10.0.0.3 pid=4532>  
<Host h4: h4-eth0:10.0.0.4 pid=4534>  
<Host h5: h5-eth0:10.0.0.5 pid=4536>  
<Host h6: h6-eth0:10.0.0.6 pid=4538>  
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=4543>  
<OVSSwitch s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=4546>  
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None,s3-eth3:None pid=4549>  
<OVSSwitch s4: lo:127.0.0.1,s4-eth1:None,s4-eth2:None,s4-eth3:None pid=4552>  
<OVSSwitch s5: lo:127.0.0.1,s5-eth1:None,s5-eth2:None,s5-eth3:None pid=4555>  
<OVSSwitch s6: lo:127.0.0.1,s6-eth1:None,s6-eth2:None pid=4558>  
<Controller c0: 127.0.0.1:6653 pid=4521>  
mininet> h1 ifconfig -a  
h1-eth0  Link encap:Ethernet  HWaddr 00:00:00:00:00:01  
          inet addr:10.0.0.1  Bcast:10.255.255.255  Mask:255.0.0.0  
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1  
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0  
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0  
          collisions:0 txqueuelen:1000  
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

mininet@mininet-vm: ~

```
lo          Link encap:Local Loopback
            inet addr:127.0.0.1  Mask:255.0.0.0
            UP LOOPBACK RUNNING  MTU:65536  Metric:1
            RX packets:0 errors:0 dropped:0 overruns:0 frame:0
            TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:0
            RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

mininet> sl ifconfig -a
eth0        Link encap:Ethernet  HWaddr 08:00:27:7e:0f:24
            inet addr:192.168.56.104  Bcast:192.168.56.255  Mask:255.255.255.0
            UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
            RX packets:1325 errors:0 dropped:0 overruns:0 frame:0
            TX packets:1128 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:1000
            RX bytes:113732 (113.7 KB)  TX bytes:159544 (159.5 KB)

eth1        Link encap:Ethernet  HWaddr 08:00:27:a7:c9:40
            inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
            UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
            RX packets:227 errors:0 dropped:0 overruns:0 frame:0
            TX packets:255 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:1000
            RX bytes:26903 (26.9 KB)  TX bytes:21747 (21.7 KB)

lo          Link encap:Local Loopback
            inet addr:127.0.0.1  Mask:255.0.0.0
            UP LOOPBACK RUNNING  MTU:65536  Metric:1
            RX packets:6841 errors:0 dropped:0 overruns:0 frame:0
            TX packets:6841 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:0
            RX bytes:433632 (433.6 KB)  TX bytes:433632 (433.6 KB)

ovs-system Link encap:Ethernet  HWaddr 92:24:0d:dc:50:59
            BROADCAST MULTICAST  MTU:1500  Metric:1
            RX packets:0 errors:0 dropped:0 overruns:0 frame:0
            TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
            collisions:0 txqueuelen:0
            RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

```
s1      Link encap:Ethernet  HWaddr a2:20:23:29:b0:4c
        UP BROADCAST RUNNING MTU:1500 Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s2      Link encap:Ethernet  HWaddr 86:dc:5e:ce:d1:4f
        UP BROADCAST RUNNING MTU:1500 Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s3      Link encap:Ethernet  HWaddr 46:a2:07:ce:9f:43
        UP BROADCAST RUNNING MTU:1500 Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s4      Link encap:Ethernet  HWaddr 1a:9f:73:b9:3e:42
        UP BROADCAST RUNNING MTU:1500 Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s5      Link encap:Ethernet  HWaddr 86:c6:1a:5e:88:49
        UP BROADCAST RUNNING MTU:1500 Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s6      Link encap:Ethernet  HWaddr 72:c6:dc:0a:fe:49
        UP BROADCAST RUNNING MTU:1500 Metric:1
        RX packets:0 errors:0 dropped:0 overruns:0 frame:0
        TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

 mininet@mininet-vm: ~

```
s1-eth1  Link encap:Ethernet  HWaddr 46:27:ff:93:59:b3
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s1-eth2  Link encap:Ethernet  HWaddr 0a:2d:f7:ef:97:d7
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s2-eth1  Link encap:Ethernet  HWaddr 32:a6:ee:6f:f7:98
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s2-eth2  Link encap:Ethernet  HWaddr aa:2c:04:68:d9:3c
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s2-eth3  Link encap:Ethernet  HWaddr ae:d0:bf:6c:f5:43
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s3-eth1  Link encap:Ethernet  HWaddr 9e:28:67:79:ef:d9
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

```
s3-eth3   Link encap:Ethernet  HWaddr 66:d9:b5:a6:cf:5a
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s4-eth1   Link encap:Ethernet  HWaddr 3a:45:5d:bb:7c:41
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s4-eth2   Link encap:Ethernet  HWaddr 5a:50:23:94:5a:92
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s4-eth3   Link encap:Ethernet  HWaddr 3e:5c:f6:0e:34:09
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s5-eth1   Link encap:Ethernet  HWaddr 0e:26:2b:fa:e7:2b
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s5-eth2   Link encap:Ethernet  HWaddr 62:51:a6:f6:c4:2f
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

```

mininet@mininet-vm: ~
    collisions:0 txqueuelen:1000
    RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s5-eth3  Link encap:Ethernet  HWaddr 6a:5b:06:0f:51:85
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s6-eth1  Link encap:Ethernet  HWaddr 76:31:94:7b:f7:3b
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

s6-eth2  Link encap:Ethernet  HWaddr fa:f8:da:fd:ef:b0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

```

TESTE DE CONECTIVIDADE

Testar conectividade entre todos os nós: `pingall`

Ping específico entre dois hosts: `h1 ping -c 4 h6`

```

mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6
h2 -> h1 h3 h4 h5 h6
h3 -> h1 h2 h4 h5 h6
h4 -> h1 h2 h3 h5 h6
h5 -> h1 h2 h3 h4 h6
h6 -> h1 h2 h3 h4 h5
*** Results: 0% dropped (30/30 received)
mininet> h1 ping -c 4 h6
PING 10.0.0.6 (10.0.0.6) 56(84) bytes of data.
64 bytes from 10.0.0.6: icmp_seq=1 ttl=64 time=0.469 ms
64 bytes from 10.0.0.6: icmp_seq=2 ttl=64 time=0.115 ms
64 bytes from 10.0.0.6: icmp_seq=3 ttl=64 time=0.252 ms
64 bytes from 10.0.0.6: icmp_seq=4 ttl=64 time=0.124 ms

--- 10.0.0.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 0.115/0.240/0.469/0.142 ms

```

TESTE DE DESEMPENHO

Abra terminais separados para h1 e h2:

xterm h1

xterm h2

No terminal do h1 (servidor): `iperf -s -p 5555 -i 1`

No terminal do h2 (cliente): `iperf -c 10.0.0.1 -p 5555 -i 1 -t 15`

```
"Node: h1"
root@mininet-vms:~# iperf -s -p 5555 -i 1

-----
Server listening on TCP port 5555
TCP window size: 85,3 KByte (default)
-----
[ 32] local 10.0.0.1 port 5555 connected with 10.0.0.2 port 38148
[ ID] Interval      Transfer    Bandwidth
[ 32] 0.0- 1.0 sec  2.83 MBytes 23,8 Mbits/sec
[ 32] 1.0- 2.0 sec  2.82 MBytes 23,6 Mbits/sec
[ 32] 2.0- 3.0 sec  2.82 MBytes 23,6 Mbits/sec
[ 32] 3.0- 4.0 sec  2.81 MBytes 23,6 Mbits/sec
[ 32] 4.0- 5.0 sec  2.82 MBytes 23,6 Mbits/sec
[ 32] 5.0- 6.0 sec  2.82 MBytes 23,6 Mbits/sec
[ 32] 6.0- 7.0 sec  2.84 MBytes 23,8 Mbits/sec
[ 32] 7.0- 8.0 sec  2.82 MBytes 23,7 Mbits/sec
[ 32] 8.0- 9.0 sec  2.81 MBytes 23,6 Mbits/sec
[ 32] 9.0-10.0 sec  2.82 MBytes 23,6 Mbits/sec
[ 32] 10.0-11.0 sec 2.84 MBytes 23,8 Mbits/sec
[ 32] 11.0-12.0 sec 2.84 MBytes 23,8 Mbits/sec
[ 32] 12.0-13.0 sec 2.81 MBytes 23,6 Mbits/sec
[ 32] 13.0-14.0 sec 2.82 MBytes 23,6 Mbits/sec
[ 32] 14.0-15.0 sec 2.82 MBytes 23,7 Mbits/sec
[ 32] 0.0-15.0 sec 43,8 MBytes 23,7 Mbits/sec
[ 32]

-----
"Node: h2"
root@mininet-vms:~# iperf -c 10.0.0.1 -p 5555 -i 1 -t 15

-----
Client connecting to 10.0.0.1, TCP port 5555
TCP window size: 85,3 KByte (default)
-----
[ 31] local 10.0.0.2 port 38148 connected with 10.0.0.1 port 5555
[ ID] Interval      Transfer    Bandwidth
[ 31] 0.0- 1.0 sec  4.12 MBytes 34,6 Mbits/sec
[ 31] 1.0- 2.0 sec  2.62 MBytes 22,0 Mbits/sec
[ 31] 2.0- 3.0 sec  3.00 MBytes 25,2 Mbits/sec
[ 31] 3.0- 4.0 sec  2.62 MBytes 22,0 Mbits/sec
[ 31] 4.0- 5.0 sec  3.12 MBytes 26,2 Mbits/sec
[ 31] 5.0- 6.0 sec  2.62 MBytes 22,0 Mbits/sec
[ 31] 6.0- 7.0 sec  2.62 MBytes 22,0 Mbits/sec
[ 31] 7.0- 8.0 sec  3.00 MBytes 25,2 Mbits/sec
[ 31] 8.0- 9.0 sec  2.62 MBytes 22,0 Mbits/sec
[ 31] 9.0-10.0 sec  2.75 MBytes 23,1 Mbits/sec
[ 31] 10.0-11.0 sec  3.25 MBytes 27,3 Mbits/sec
[ 31] 11.0-12.0 sec  2.88 MBytes 24,1 Mbits/sec
[ 31] 12.0-13.0 sec  2.62 MBytes 22,0 Mbits/sec
[ 31] 13.0-14.0 sec  3.00 MBytes 25,2 Mbits/sec
[ 31] 14.0-15.0 sec  2.75 MBytes 23,1 Mbits/sec
[ 31] 0.0-15.0 sec 43,8 MBytes 24,2 Mbits/sec
[ 31]
root@mininet-vms:~#
```